

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool: Viva Voce

Weightage: 10%

QUESTIONS:

Total Marks: 20

1. Define SOAP.
2. Define WSDL.
3. What is an API?
4. Define XML Schema.
5. What is a Web Service?
6. Define REST.
7. What is a Protocol?
8. Define Endpoint in web services.
9. What is a Service in web applications?
10. What is the difference between SOAP and REST?
11. What is the role of WSDL in web services?
12. What is JSON and how is it used in APIs?
13. What is HTTP and why is it used in web services?
14. What is meant by request and response in web communication?
15. What is the importance of XML in web services?
16. What is the purpose of an API endpoint?
17. What is the difference between HTTP and HTTPS?
18. What is meant by RESTful services?
19. What is a namespace in XML?
20. What is the role of a client and server in web services?

Code of Conduct:

*I N.Loka Bandhava/192324167____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively

1. Define SOAP.

Answer: SOAP stands for **Simple Object Access Protocol**. It is an XML-based protocol used for exchanging structured information between applications over a network.

2. Define WSDL.

Answer: WSDL stands for **Web Services Description Language**. It describes a web service, including its operations, messages, data types, and endpoint.

3. What is an API?

Answer: API stands for **Application Programming Interface**. It allows two software applications to communicate and exchange data with each other.

4. Define XML Schema.

Answer: XML Schema defines the **structure, elements, data types, and rules** of an XML document. It is used to validate whether XML data is correctly formatted.

5. What is a Web Service?

Answer: A web service is a software component that allows different applications to **communicate over a network**. It enables data exchange between different platforms and technologies.

6. Define REST.

Answer: REST stands for **Representational State Transfer**. It is an architectural style that uses HTTP methods such as **GET, POST, PUT, and DELETE** to access resources.

7. What is a Protocol?

Answer: A protocol is a **set of rules** that defines how data is transmitted and received between devices or applications. Examples include HTTP, HTTPS, and TCP/IP.

8. Define Endpoint in web services.

Answer: An endpoint is a **specific URL or address** where a web service can be accessed. It allows clients to send requests to a particular service.

9. What is a Service in web applications?

Answer: A service is a software function that performs a **specific task** and provides functionality to other applications. It usually communicates through APIs or web services.

10. What is the difference between SOAP and REST?

Answer: SOAP is a **protocol** that mainly uses XML, while **REST is an architectural style** that commonly uses JSON or XML. REST is generally simpler and more lightweight than SOAP.

11. What is the role of WSDL in web services?

Answer: WSDL provides a **formal description of a web service**. It tells the client what operations are available, what data is required, and where the service is located.

12. What is JSON and how is it used in APIs?

Answer: JSON stands for **JavaScript Object Notation**. It is a lightweight data format commonly used by APIs to exchange data between clients and servers.

13. What is HTTP and why is it used in web services?

Answer: HTTP stands for **HyperText Transfer Protocol**. It is used to send requests and responses between clients and servers over the web.

14. What is meant by request and response in web communication?

Answer: A **request** is a message sent by the client to the server asking for a resource or operation. A **response** is the server's reply containing the requested data or result.

15. What is the importance of XML in web services?

Answer: XML provides a **structured and platform-independent way** to represent and exchange data. It is especially important in SOAP-based web services.

16. What is the purpose of an API endpoint?

Answer: An API endpoint provides a **specific access point** for a particular API resource or operation. Clients use its URL to send requests and receive responses.

17. What is the difference between HTTP and HTTPS?

Answer: HTTP transfers data without encryption, while **HTTPS uses encryption through TLS/SSL**. HTTPS provides better security for sensitive information.

18. What is meant by RESTful services?

Answer: RESTful services are web services designed according to **REST principles**. They use resources, URLs, and standard HTTP methods such as GET, POST, PUT, and DELETE.

19. What is a namespace in XML?

Answer: An XML namespace is used to **uniquely identify elements and attributes** in an XML document. It prevents naming conflicts when different XML vocabularies are used together.

20. What is the role of a client and server in web services?

Answer: The **client sends a request** for a service or data, while the **server processes the request and sends a response**. This enables communication between applications